SomaTek AI

BSc. in Software Engineering

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CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Official projections show that Africa' population will undergo rapid growth. For example, the IMF's analysis demonstrates that "by 2050, Africa's population will reach close to 2.5 billion" (Stanley, 2025). Rwanda, like many African countries, has placed human capital development at the center of its National Strategy for Transformation (Mungai, van der Linden, & Munezero, 2025). Similarly, the World Bank has reported Rwanda's country memorandum strengthening human capital through health, education and skills (World Bank 2024). As a result, Rwanda is at a crucial junction to increase the learning outcomes obtained from their education system. High learning outcomes are directly proportional to the strengthening of human capital which in return will promote a boost in sustainable economic growth.

Rwanda has made astounding progress in school enrollment and education system reforms where in 2023/2024 it met its National Strategy for Transformation goal for improving enrolments by 41% from 2020. The new target for 2030 was set to 65% (MINEDUC, 2025). However, learning quality remains a persistent challenge. UNESCO defined the term learning poverty as the amount of 10 year olds that are unable to read and understand simple text. It combines poor reading proficiency and out of school rates. The World Bank in its analysis to end learning poverty reported before COVID-19, about 57% of children in low- and middle-income countries could not read by the end of primary, rising to 86% in Sub-Saharan Africa.(World Bank, 2022). This poses a major threat to the continent's ability to produce the skilled labourers needed for different industries like agriculture, technology, health, education and industry.

Rwanda being a Belgian colony, French was the official language of administration and schooling for multiple decades. Once Rwanda joined the East African Community in the year 2008, it adopted English as the principal language of instruction across the education system (UNICEF, 2009). While this transition was politically strategic. The education system was heavily impacted. A UNESCO analysis titled "All children in Rwanda are born to learn" found only about 20% of Rwandan primary pupils reach minimum reading proficiency(UNESCO, 2014). Dr John Rutayisire, director of the Rwanda Education Board, defended the abruptness of the policy that decreed that teachers would start using English within less than four months (The Guardian, 2012).

A study found that while students like to study English as a subject, many performed badly and teachers themselves felt they needed more training in English (Nzabimfura & Andala, 2022). Additionally, a study conducted in the Rubavu district demonstrated that students that learn exclusively in English often perform worse academically than those taught using a combination of English and Kinywarwanda (Mahoro, Ngabonziza 2024). The findings indicate that instructional communication and code-switching, rather than the curriculum itself, play a decisive role in learner comprehension and academic performance. Instructors who are able to mediate content in a diverse linguistic manner through rephrasing and translation and contextualizing explanations actually improve learners' understanding.

Rwanda in response to the learning quality challenges has started integration standardized teaching approaches. The RwandaEQUIP aims to improve consistency and reduce variation in instructional delivery by providing teachers with predefined curriculums and lesson plans in English. Their pedagogical strategy emphasizes the main instructional language to be English. While these strategies are going to improve consistency and reduce variation in teaching, they assume a level of teacher and student language proficiency that is not often present. As a result, standardized lessons may be offered but not understood, hence the learning poverty will remain..

Technological advancements particularly in Artificial Intelligence have offered new opportunities to fix this gap. Rather than replacing teachers or curricula, AI can be used to improve teacher's instructional capacity. This specifically in linguistic complex environments. This technology could also be available to students as a means of getting contextualized instructional content and hence improving learning outcomes.

This study proposes SomaTek AI as a system that is designed to support instructional language mediation. This system seeks to enhance the facilitator's ability to explain, adapt and contextualize instructional content. Therefore contributing to long-term human capital development.

1.2 Problem statement

Despite significant efforts in education reforms and the adoption of standardized pedagogy, learning outcomes in Rwanda continue to be compromised primarily due to persistent challenges in the language of instruction. These challenges are particularly pronounced in contexts where English, though designated as the primary instructional language, is not the first language for either teachers or learners. Empirical evidence from Rwandan classrooms demonstrates that students' comprehension and academic performance are closely tied to the effectiveness of explanation, rephrasing, and the language used during instruction. While standardized lesson plans have improved structural consistency, they have failed to address the critical issue of widespread low English proficiency among both teachers and students, which restricts effective real-time instructional communication. Moreover, existing student-facing AI large language models provide explanations exclusively in English, lack contextual adaptation to the educational environment, and do not allow students to revisit previous explanations. These tools also remain disconnected from the teacher's instructional intent, leading to parallel but misaligned learning pathways that can undermine the goals of standardized pedagogy. This situation reveals a clear and pressing research gap: there is an urgent need for an AI-mediated system specifically designed to facilitate shared, adaptive, and multilingual instructional communication between teachers and students, thereby aligning classroom teaching with genuine learner comprehension.

1.3 Project's main objective

The main objective of this project is to design, develop, and evaluate SomaTek AI as a student-teacher focused instructional mediation system that enhances shared understanding through adaptive, multilingual explanations. Specifically, the project aims to increase student comprehension of taught concepts by at least 20%, as measured by standardized assessments, within multilingual and low-English proficiency educational contexts. The system will

provide real-time and revisitable explanations to address linguistic and communication gaps, with at least 80% of participating teachers and students reporting improved instructional communication. By strengthening instructional communication rather than replacing existing pedagogical structures, the project intends to demonstrably improve learning quality and support measurable gains in learner achievement and human capital development.

1. Investigating communication challenges in multilingual classrooms by reviewing existing solutions and collecting data on language of instruction, teacher explanation strategies, and lastly student comprehension in Rwandan schools.
2. To design and develop SomaTek AI a student-teacher focused instructional mediation system that leverages Natural Language Processing techniques to generate adaptive, multilingual explanations that align teacher instructions with student comprehension.
3. Evaluating effectiveness of the proposed system using problem-specific, measurable indicators, that include improvements in student comprehension of taught concepts

1.4 Research questions

1. What instructional communication challenges do teachers and students face in multilingual classrooms, specifically where English is the primary language of instruction.
2. How can an AI powered tool be built to support shared instructional communication between teachers and students through adaptive and multilingual explanations ?
3. At which level does the use of SomaTek AI improve student comprehension and alignment between teacher explanations and student understanding?
4. How do SomaTek's primary users perceive the usability and instructional value of an AI-assisted explanation system in classroom and learning contexts?

1.5 Project scope

This project's main focus is the design, implementation and evaluation of SomaTek AI as a student-teacher focused AI-powered instructional mediation tool that supports instructional communication across multiple levels of education. In more detail the scope includes:

1. Secondary school and university-level education, with emphasis on theory-intensive and language-demanding subjects.
2. Support for advanced paraphrasing, detailed explanation, and multi-lingual code switching.
3. Learning and instructional analytics to support teachers and educational stakeholders by offering aggregated and anonymous feedback in learner engagement and comprehension.

1.6 Significance and Justification

Upon successful completion of this project, Somatek AI will enable educators and learners to overcome the challenges that result from low English proficiency. By supporting comprehension and production of academic English, the system will enable learners and educators to engage more effectively with the course materials, classroom explanations and assessments thereby improving overall learning quality.

This project will also provide both teachers and education stakeholders with valuable data into comprehension related learning challenges. Through advanced analytics on comprehension patterns and language difficulty, institutions can better understand where learners and teachers struggle and make informed decisions regarding instructional support, curriculum design and targeted interventions.

In the long term, SomaTek AI has the ability to build sustainable and impactful improvements in education quality within multilingual and under-resourced contexts.

1.7 Research Budget

Item	Unit Price	Quantity	Total
Google Colab Pro (compute)	$9.99 / month	3 months	$29.97
Cloud Hosting Service	$10.00 / month	3 months	$30.00
Domain Name (web address)	$12.00	1 year	$12.00
API Subscription/Usage	$10.00 / 3 mo	1 3-month term	$10.00
Total Estimated Cost			$81.97

Fig 1. Research Budget

1.8 Research Timeline

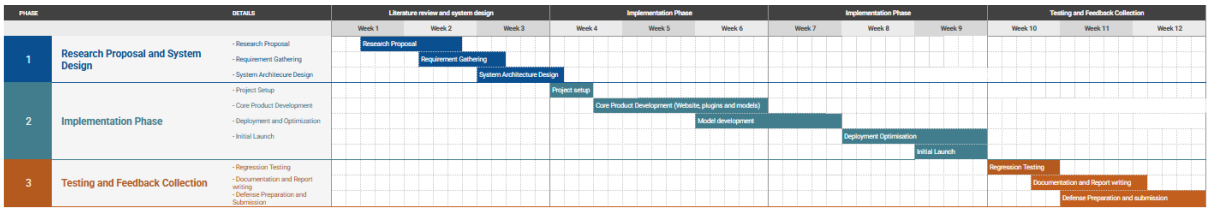

Fig 2. Gantt Chart

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter presents assessment of academic and technical literature on quality of learning outcomes. Over the last years Rwanda has made astounding progress in improving access to education. UNESCO reports Rwanda implemented "fee-free basic education" among key reforms (UNESCO 2022). Moreover Education access was improved through more policies like school feeding programs and increased teacher recruitment. This has significantly increased enrollment and completion rates across the country. These efforts have contributed to a steady rise in primary school participation over the last half-century whereby currently UNICEF reported an adjusted net attendance rate for children of primary school age of 95% (UNICEF, 2025). Despite the improvements in access to education, concerns regarding the quality of learning outcomes persist.

Over 80% learners in Sub-Saharan Africa complete several years of schooling without acquiring the basic literacy and numerical skills, learners cannot read or understand a simple 150-word text by age 10, according to the UN. UNESCO statistics indicate students in Rwanda score 358 on a harmonized test score scale where 625 represents advanced attainment and 300 represents minimum attainment (World Bank, 2024). This concept is what is referred to as the learning crisis in Africa, this is characterized by poor foundational skills, weak comprehension and limited ability to apply knowledge. The crisis is complex and multifaceted, it is shaped by structural barriers, pedagogical limitations and system challenges linked to the language of instruction and how lessons are delivered.

By comparison, well resourced schools in other countries average a significantly higher OECD mean in reading math. OECD's 2022 PISA have shown Swedish 15 year olds average 73% in reading and math proficiency versus 69% OECD-wide (PISA, 2022). This disparity shows that an improvement to access won't necessarily affect learning outcomes.

2.2 Historical Background of the Research Topic

Rwanda is a Monolingual nation, majority of citizens speak the national language Kinyarwanda. In the past, Rwanda's education was largely francophone since 1962. However, in 2009, the language was abruptly changed to English. French and Kinyarwanda was the dominant language of instruction for several decades. The training institutions for teachers were structured in the same way (Samuelson & Freedman, 2010). In terms of quality it is needless to say that this was an immense shock to the education system. This mismatch led to marked changes in student outcomes: pupils tested in Kinyarwanda scored far higher (64% meeting benchmarks) than those tested in English (only 10% meeting benchmarks) after the switch. Recurrent policy changes (e.g 2009 - 2020) and limited teacher training in English consistently causes a loss of focus on English instruction and hurts learning in other subjects as well.

More reforms contributed to the challenges, a second reform, The Competence Based Curriculum that rolled out between 2015 and 2016 and first examined nationally at the O-Level in 2018 (MINEDUC, 2015). This reform encourages learner-centered pedagogy. It was focusing on thinking skills and continuous assessments and retained English as the language of instruction. This shift added to the strain on teachers and students. The O-Level pass rate declined from approximately 90% in 2017, the final cohort under the previous

curriculum, to 83% in 2018 (Word Bank, 2020). Education analysts attributed this decline not only to the difficulty of the curriculum but also the non-ready teachers and lack of sufficient instructional support (World Bank, 2020, UNESCO 2019).

In brief, Rwanda's education system reforms illustrate that multiple large scale instructional reforms, especially when combined with language transitions, can have multiple effects on learning outcomes. Rather than a single policy shift causing decline in performance decline, it is the combination between the language of instruction shift and curriculum changes that has persistently contributed to comprehension and uneven student achievements.

2.3 Overview of Existing System

In Sub-Saharan Africa, there is a growing set of education platforms that target teacher-student communications in different languages. Some examples include the following:

- Luma Learn: This is a Whatsapp based AI tutoring platform for Grades 8-12. Luma provides on-demand Q&A, quizzes and exam prep in 11 official languages. It delivers curriculum aligned literacy and numeracy practice. It also reduces barriers by using Whatsapp and involves schools in the design process to ensure relevance. This however is not applicable to children living in low-internet places.
- M-Shule: An AI-driven SMS/Whatsapp learning platform. The M-Shule's "LEAD" system lets teachers send personalized lessons and quizzes via basic phone SMS or Whatsapp, using simple language learning activities. It supports multiple Kenyan languages (e.g. English, Kiswahili, Kikuyu, ...) this provides a teacher portal for monitory learning and engagement. M-Shule's strength is high scalability to low-income users; its limitations are sending multimedia content through SMS.
- Kytabu(Kenya): A suite of AI tools for teachers and students. Kytabu's Hodari app aims to assist teachers with grading, lesson planning and administrative tasks. It also designed a hands-free teacher support tool in the form of smart glasses called "Hodari Vision". These are all non-scalable solutions in the Rwandan context due to the high price of wearable technology. Or just the fact that it actually revolves around lesson planning which has already been taken care of by RwandaEQUIP.
- Other Software platforms offer translation, adaptive content only if prompted well enough. These include ChatGPT, GeminiAI and more. These are not fair to low proficiency learners that are not able to prompt well.

2.4 Review of Related Work

Recent research and reports show the promise and challenges of AI in multilingual, adaptive education. A survey of the described educational multilingual AI tutors as systems that combine LLMs, multimodal inputs and pedagogical techniques to deliver personalized instructions across languages. These systems leverage techniques like retrieval-augmented generation and knowledge tracing to align AI output with curriculum content. The above tools have proven with controlled trials that students learn significantly more in less time when they are equipped with a well designed AI tutor than with traditional active learning, and feel more engaged. In language learning, AI-mediated instruction has likely been found to improve outcomes and motivation: i.e a study reported that AI translation and feedback significantly improved english learning outcomes, increased motivation and fostered more

self-regulated learning behaviors among ELL students. Case studies of systems like SingaKids have demonstrated that AI tutors can provide effective dialogic teaching in multiple languages.

2.4.1 Summary of Reviewed Literature

Briefly, the reviewed studies and reports show that AI driven adaptive tutors and translation have lately been increasingly applied to language education, they can remarkably improve student engagement and achievement when aligned with proper pedagogy. Platforms like ChatGPT, Perplexity and integrated LMS use AI to generate multilingual lesson content and examples but common flaws include context errors in translation, no pedagogical structure in general-purpose AI, and it's unfairly hard for learners with low proficiency to prompt. These findings suggest that while AI has great potential, there remains a gap for systems specifically designed to mediate teacher-student communication and adapt explanations in real classroom settings.

2.5 Strengths and Weaknesses of the Existing System(s)

The already existing solutions offer notably personalized learning at a large scale. They all have an impact on the learning outcomes of their respective users. However they also have limitations as mentioned below:

1. Luma Learn: This tool is widely accessible through Whatsapp and it supports multiple languages, it's also curriculum-aligned. However, it has limited effectiveness in low-internet-areas; It mainly relies on student motivation and connectivity
2. M-Shule: This tool is very scalable, it works on basic phones, it also offers personalized lessons. However this is not able to deliver visual content and it's restricted to text based communication (short & no visuals)
3. Kytabu: This supports the facilitator's efficiency; it reduces administrative work and allows teachers to focus on important tasks. It offers hands-free assistance via Hodari Vision. This however is not cost-effective due to the price of wearable technology.
4. Other Software platforms like ChatGPT , Gemini and more. Can translate and simplify content down to the level of any student, they are also widely available with 24/7 support. These tools however are not scoped to the context of the student and teacher. These are also at high risk of stating wrong information confidently.

Despite the fact that these tools exist and are fixing a critical problem. Standalone translators or general LLM chatbots lack pedagogical context awareness. AI translations can be literal and sometimes miss cultural or contextual nuance, this confuses students even more. More broadly, LLMs are more prone to confidently state the wrong information if not carefully constrained. Privacy and bias are further concerns when using student data with AI. In brief, while current systems can make content more accessible over the internet, they tend to focus on either translation or content generation, just a few provide a seamless "mediator" that contextualizes teacher explanations for students whose native language is not English.

2.6 Research Gap

Despite the rapid growth of Ed-Tech tools in Africa and the world. There still exists several gaps, especially in the context of learning outcomes:

1. Insufficient context of AI powered tools: The AI powered are meant for general purpose and don't have pedagogical intent when simplifying or providing translations. This is a limiting factor to aligning teacher explanations (Luckin et al., 2016; Zawacki-Richter et al., 2019).
2. Accessibility: Platforms like Kytabu and Hodari Vision are not cost-effective in low-resource contexts like Rwanda. Same thing with Whatsapp, or SMS-based systems. These require multiple devices as well.(Ma et al., 2014).
3. Prompting: These tools require to be prompted which is not fair for people that can merely speak English. They need to be efficiently prompted to return actually good explanations. This is a barrier to students with low English proficiency (Samuelson & Freedman, 2010; UNESCO, 2019.

   In summary, the literature highlights a deep gap. Existing tools either focus on student-centered application, teacher productivity or content translation. This fails to provide common understanding in the classroom. It doesn't address the gap enhancing effectiveness or learning outcomes but rather promotes students to using AI irresponsibly. The development of SomaTek AI will aim to enhance both teacher efficiency and student comprehension through AI mediated explanations.

2.6 General comment and Conclusion

Briefly, different software tools already aim to support instruction in multilingual places, from translator, transcribers and captioning apps to adaptive tutoring learning management systems and LLM chatbots. All these address part of the problem. Worldly and Interprenet translate spoken classes into dozens of languages, Kytabu, Luma Learn adaptively transform content to the learner's needs. ChatGPT, GeminiAI can re-express materials in simpler or translated forms. However, none of these solutions fully fits in the teacher-student interaction dynamics. As education experts suggest, AI should be a powerful ally rather than a replacement for human educators.

As a result, the current systems tend to either focus on student adaptation or just provide English content that is standard without addressing the language proficiency constraint in both teachers and learners. This creates a gap where instructions delivered remain linguistically incomprehensible despite the support given by technology. The proposed SomaTek AI aims to fill the gap by functioning as an AI-powered English instructional aid that supports both teachers and students in comprehension and interaction with instructional content in English. Rather than just mediating teacher explanations, it enables teachers to be more effective while teaching and learn much faster regardless of individual English proficiency levels.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter provides an overview of the analysis and design of an AI powered educational system to enhance the learning outcomes of students with low English proficiency. The platform is a multi-tenant solution with transcription, analytics and querying services.

3.2 Research Design (including the development model used)

This study uses a mixed approach that combines both qualitative and quantitative approaches. The mixed approach is justified by the research problem because it involves both measurable learning outcomes and contextual instructional communication processes. The methods that are quantitative allow objective measurement of changes in student comprehension and instructional alignment before and after the implementation of the solution. Qualitative methods however provide more detail about the reason why these changes occur in multilingual classrooms.

3.2.1 Qualitative Research

The qualitative method of this study will be focused on understanding instructional communication challenges and user experiences associated with the use of SomaTek AI. Data collection will be done through interviews that are semi-structured, live classroom observations and open ended questionnaires that will be administered to both teachers and students.

Interviews with teachers will explore the different pedagogical approaches, impact on lesson delivery and changes in student understanding. Student feedback will focus on usability, clarity of explanations and usefulness of features such as multilingual support and explanation replay.

3.2.2 Quantitative Research

The quantitative component of the study aims to measure the effectiveness of SomaTek AI in improving student's understanding and curriculum alignment. Quantitative data will be collected using pre-test and post-test assessments designed to evaluate student's understanding of concepts taught in classrooms. These assessments will be in line with the content shared class and administered before and after the introduction of the system.

The system will itself generate quantitative metrics that will include system-generated analytics such as student engagement, quiz marks and search topics. These statistics when analysed will be used to evaluate performance and engagement over time. These measurements provide objective evidence of how the system contributes to measurable improvement in learning outcomes

3.2.3 Sampling Strategy

The sampling strategy for the research is mainly the target population. These are students and teachers in secondary schools and universities, teachers that teach theory heavy subjects English and education settings where multilingual instruction and low-to-moderate English proficiency.

The sampling technique to be used will be purposive sampling, this will involve selecting participants that best represent the linguistic and instructional barriers. The schools to be selected shall have English as their official language, multilingual school-goers and the willingness to integrate SomaTekAI into their workflows.

The sample size that is feasible with the project's hosting budget will be 2 schools and less than 100 users. Among the users 2 teachers, 98 students. This sample size is enough to detect meaningful progress and collect various data patterns.

3.2.4 Ethical Considerations

Ethical approval will be requested from relevant education authorities. Consent will be secured from all participants, in case of minors parental consent will be obtained for minors. Data will be anonymized, securely stored, and used solely for academic research purposes.

3.4 Development Model

This research project will be carried out in an iterative and incremental development model. It will follow Agile methodology principles. I selected this method due to the nature of the project, there is a need for recurrent changes and refinement to system features and integration of machine learning components that perform based on iterative evaluation and adjustment.

Given the tight deadline, the development's main emphasis is to achieve a prototype that can be used and improved progressively. The development process will be done in the following phases:

- Requirement Gathering: This will involve identifying both functional and non-functional requirements of the system. After analysis of the different literatures, observations of instructional challenges and identifying teachers and student processes.

- Design: This stage will involve designing a prototype that meets the requirements gathered earlier on. This will involve drafting user journeys and interfaces.

- Implementation: This will involve the development of the designed solution.

- Integration testing: There will be testing the various components of the application aggressively and ensuring it stays stable.

- Quality Assurance: This stage will be about checking if the solution implemented full meets the requirements and designs proposed. It will also include regression testing and making sure that the solution performs as expected under heavy load.

3.3 Class Diagram

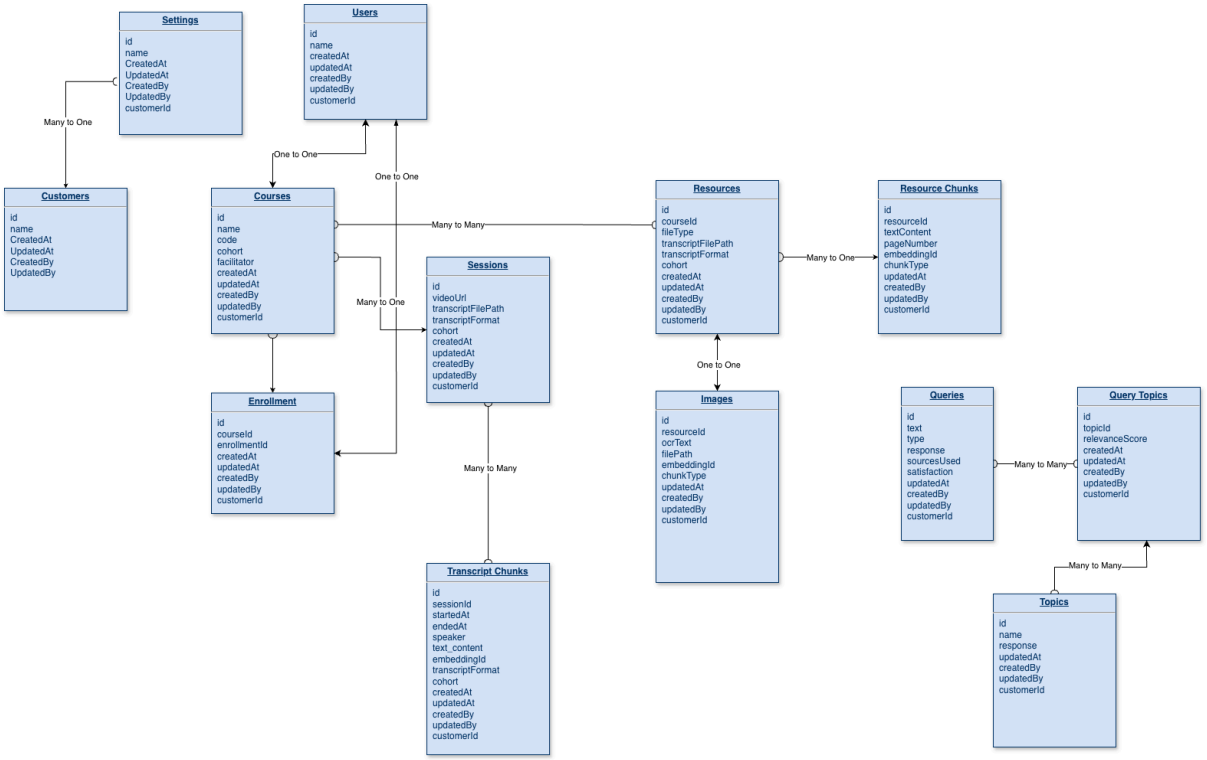

Fig 3. Entity Relationship Diagram

3.4 System Architecture

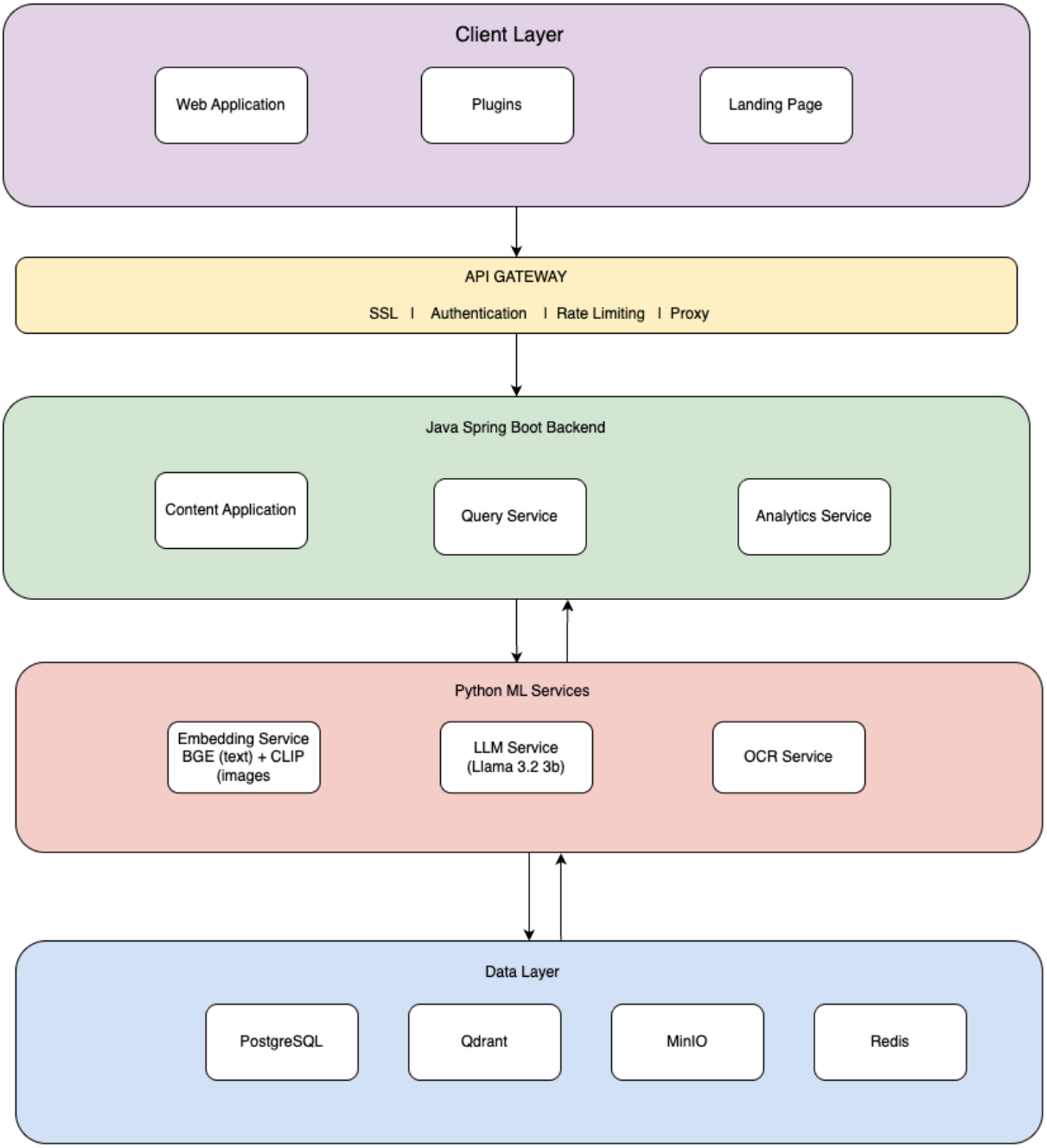

Fig 4. System Architecture

References (APA Format)

1. McGreal, C. (2008, October 14). *Rwanda to switch from French to English in schools*. *The Guardian*.

2. Stanley, A. (2023). *African Century*. IMF Finance & Development, September 2023.

3. Sinha, S., & Getachew, M. (2024, July 12). *As Africa's population crosses 1.5 billion, the demographic window is opening*. UN Economic Commission for Africa.

4. Mungai, K., van der Linden, A., & Munezero, A. (2025, April 1). *Aligning workforce skills and economic ambition in Rwanda*. Cenfri.

5. World Bank. (2024, Nov. 13). *New report outlines pathways to sustainable growth in Rwanda*. World Bank press release.

6. World Bank. (2021, July 1). *Learning poverty measure* [Brief].

7. World Bank. (2022, Oct. 12). *Ending learning poverty* [Brief].

8. UNICEF East & Southern Africa. (2018). *The impact of language policy and practice on children's learning in Rwanda* (Study report).

9. Nzabimfura, G., & Andala, H. O. (2022). *English language as a medium of instruction and learners' academic performance in secondary schools of Rwanda*. International Journal of Scientific & Research Publications, 12(5), 305.

10. UNESCO. (2023). *All children in Rwanda are born to learn: Spotlight on basic education completion and foundational learning in Rwanda*. UNESCO.

11. Abaitua, J. (2024, Dec. 2). *Rwanda advances EdTech integration but stresses need for local content*. Education Curated.
12. Clover, J. (2012, November 13). Jury out on language-switch trend. The Guardian. Retrieved from [The Guardian archives]
13. MINEDUC. (2015). *Competency-based curriculum: Curriculum framework for pre-primary, primary and secondary education*. Ministry of Education, Rwanda.

14. World Bank. (2023). *Quality basic education for human capital development project – Rwanda* (Harmonized test score data). World Bank Group.

15. OECD. (2023). *PISA 2022 results (Volume I): The state of learning and equity in education*. Organisation for Economic Co-operation and Development.

16. Lewis, B. L. Samuelson & Freedman, S. W. (2010). Language policy, multilingual education, and power in Rwanda. Language Policy, 9(2), 191–215

17. Luma Learn. (n.d.). Free WhatsApp-based AI tutoring for learners and teachers. Retrieved from https://lumalearn.ai/

18. M-Shule. (n.d.). *About us: Adaptive SMS-based learning platform for Africa*. Retrieved from https://www.mshule.com/about-us (description of the M-Shule SMS learning platform and adaptive content)
19. Kytabu Africa. (n.d.). *AI-powered educational solutions for learners and teachers*. Retrieved from https://kytabu.africa/ (information on Kytabu's tools including Hodari and Somanasi)
20. Ma, W., Adesope, O. O., Nesbit, J. C., & Liu, Q. (2014). Intelligent tutoring systems and learning outcomes: A meta-analysis. Journal of Educational Psychology, 106(4), 901–918. https://doi.org/10.1037/a0037123